

CASE STUDY

WASTE-DERIVED MORTAR

Machine learning and adaptive experiments turned concrete from a demolished bridge deck into mortar with **up to 50% less cement** and strengths **above 40 MPa**, with design cycles running **nearly 10x faster** than conventional testing.

Designing Low-Cement Mortar From Demolished Concrete

The United States generates as much as 405 million tons of construction and demolition waste (CDW) every year, and much of it is landfilled. At the same time, cement remains one of the most carbon-intensive ingredients in construction. This work, carried out at the University of Arizona, shows how computational intelligence closes that loop: recover crushed concrete as a feedstock, then design mixes that use far less cement while meeting structural strength requirements.

OVERVIEW

Every candidate mix normally needs 28 days of curing before its strength is known, which makes sustainable material design slow and expensive. We combined experiments, machine learning and Bayesian optimization so each round points straight to the most promising next mixes.

An ML model predicts 28-day strength from composition and 3-day strength, shrinking each design cycle from four weeks to a few days.

THE PROJECT

The feedstock was crushed concrete from a demolished bridge deck in Arizona, chosen for its consistent structural-grade quality and freedom from non-concrete contamination. It was crushed, milled and sieved to match the gradation of conventional fine aggregate.

The goal: find mortar mixes that minimise Portland cement while keeping 28-day compressive strength at or above 25 MPa, using CDW, Class F fly ash, natural aggregate and water.

At a Glance

- **Feedstock:** crushed concrete from a demolished bridge deck, sieved between #10 and #200.
- **Binders:** Type I/II Portland cement with Class F fly ash replacing part of the cement.
- **Target:** lowest possible cement content at 25 MPa or more after 28 days.
- **Method:** XGBoost strength model feeding a Bayesian optimization loop.
- **Validation:** 50 mm cubes tested to ASTM C109 at 3 and 28 days.



Target

Cut cement content while holding 28-day strength above the 25 MPa design requirement.



Model

Predict late-age strength from early-age tests so the search no longer waits 28 days per round.



Validate

Fabricate and test the most promising mixes to confirm every prediction in the lab.



The Challenge

Recycled concrete is a promising feedstock, but it behaves differently from natural sand. Ours absorbed four times more water (9.76% vs. 2.34%) and was lighter (specific gravity 2.26 vs. 2.55), so every mix needed careful water balancing. The bigger obstacle was time. Strength is judged at 28 days, so a conventional optimization loop of just ten rounds would take 280 days.

01	Characterize the feedstock X-ray fluorescence showed 49.4% silica in the recycled concrete, a good sign for strength development.
02	Predict 28-day strength An XGBoost model trained on recycled mortar and concrete data predicts late-age strength from composition and 3-day strength (R^2 0.957).
03	Search with Bayesian optimization A Gaussian process ranks candidate mixes within practical bounds and picks the next batch most likely to cut cement and still pass.
04	Confirm in the lab The five strongest low-cement mixes were cast and tested at 28 days, confirming the model's predictions.

What It Means for Manufacturers

- Waste streams you already handle can become qualified feedstocks.
- Fewer, smarter experiments replace months of trial and error.
- Cement, cost and carbon fall together without giving up strength.

The Problem

Concrete consumes vast amounts of limestone, sand and gravel, and cement production is a major source of global CO_2 . Yet at the end of its life, that same concrete is usually discarded as waste instead of being returned to production. Recycling it into mortar offers a double win: natural resources are conserved, and carbon-intensive cement and aggregates are partly replaced. The barrier has been quality variability and the slow, costly testing needed to prove each new mix.



Faster Cycles

Early-age tests stand in for 28-day results, so each design round takes days instead of weeks, about ten times faster.



Lower Cement

Starting mixes held 478 to 796 kg/m^3 of cement. Optimized mixes halved the cement while carrying up to 350 kg/m^3 of recycled concrete.

Results

Following the model's guidance, new mixes were predicted, fabricated and tested with up to 50% less Portland cement, and several exceeded 40 MPa, well above the 25 MPa requirement. Mixes carrying up to 350 kg/m^3 of recycled concrete stayed above 30 MPa, and lab results matched the predictions.

Cement reduction

50%

vs. lowest-cement starting mix

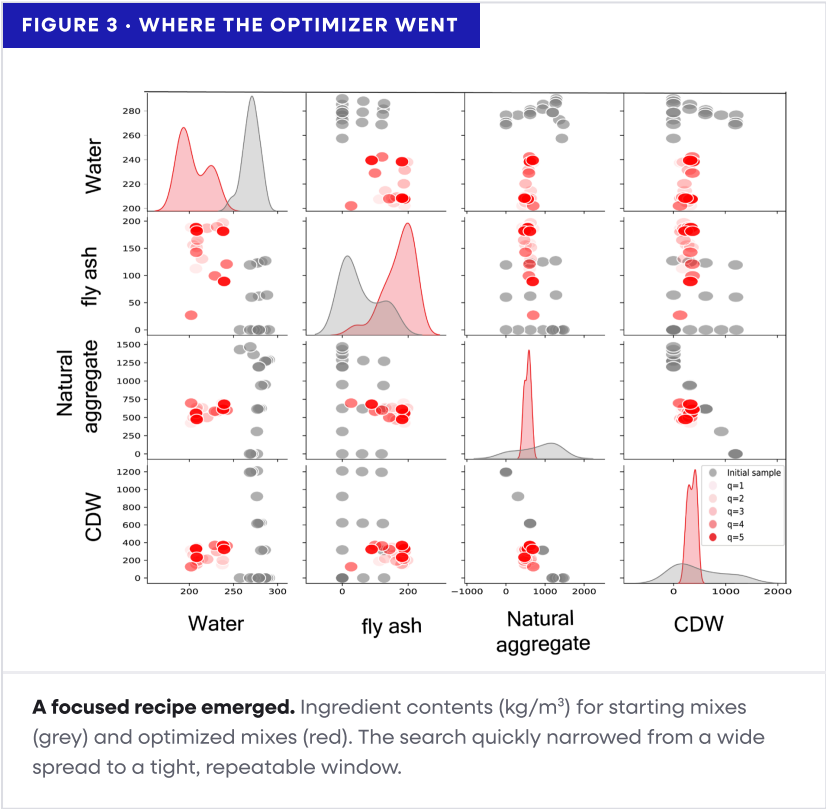
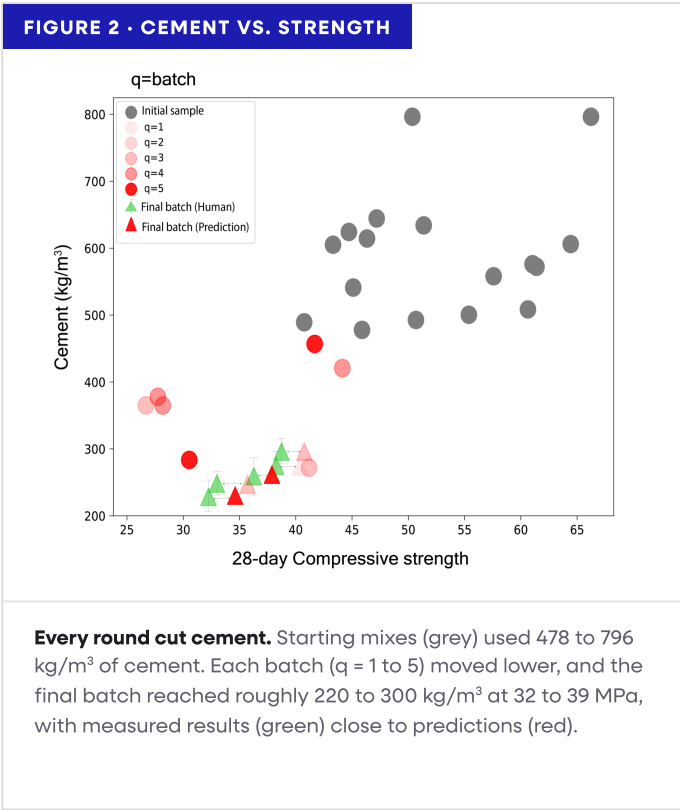
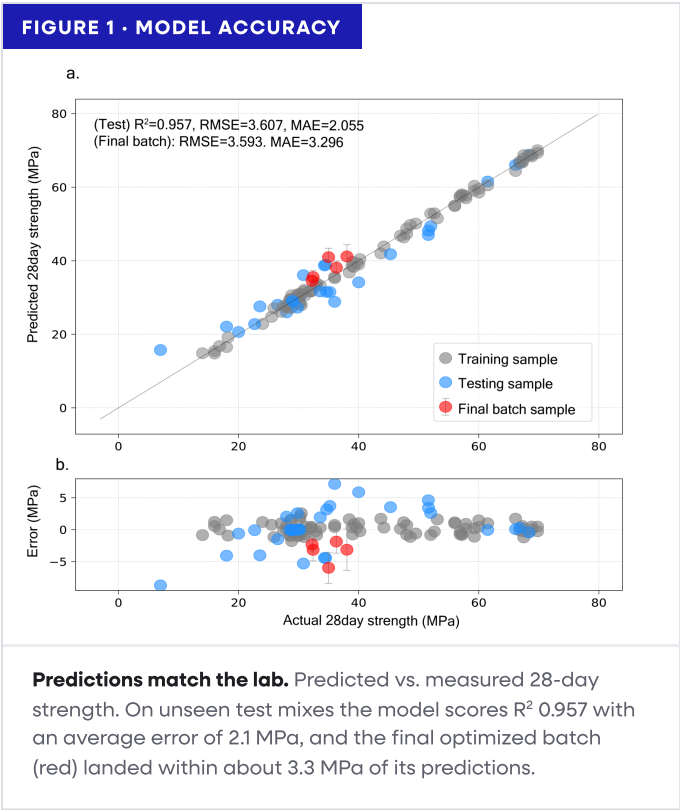
Faster design cycles

10x

3-day tests replace 28-day waits

Results in Detail

Three views of the same story: the model predicts strength accurately, each optimization round pushes cement lower, and the winning mixes share a clear recipe the team could not have found by trial and error alone.



What the Optimizer Learned

- **Less water:** about 200 to 240 kg/m³, down from 260 to 290, offsetting the recycled concrete's high absorption.
- **More fly ash:** mostly 100 to 200 kg/m³, taking over part of the cement's role.
- **Balanced aggregates:** 500 to 700 kg/m³ natural aggregate alongside 200 to 400 kg/m³ of recycled concrete.
- **Fast convergence:** five guided batches were enough to settle on the winning region.

Model accuracy on test mixes

R^2 0.957

average error 2.1 MPa